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Department of Physics

What Quantum Trace Can a Heavy Particle Leave?

Curvature–Spectator Entanglement
and the Cosmological Collider

Cumulative Research Report

Group-B

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Abstract and physical question

We calculate light–spectator Gaussian entanglement and a leading cubic spectator bispectrum in a specified two-field de Sitter model. The aim is to determine whether greater Gaussian entropy implies a stronger or more recoverable heavy-field signal. Independent oscillator, mode-evolution, canonical-commutator and in-in-integral tests establish the implementation. A mass/mixing scan compares entropy at common times. A converged squeezed bispectrum at $M = 2$, $g = 0.1$ exhibits a logarithmic clock. Mock fits test frequency recovery; physical fits show why background and window variation must supplement a narrow numerical profile. Finally, varying the cubic coupling changes the signal while leaving the quadratic entropy fixed. The result demonstrates that Gaussian entropy alone cannot determine the leading three-point amplitude.

Step 1: Model, conventions and single-field reference

We set $\hbar = c = 1$ and use $a = -1/(H\eta)$, $\eta < 0$, $x = -k\eta = k/(aH)$, $M = m/H$ and $g = \rho/H$. Numerically $H = 1$. The action is

$$S_2 = \int \mathrm{d}\eta\, \mathrm{d}^3x \Big\{ \frac{a^2}{2}[q'^2 - (\nabla q)^2 + \sigma'^2 - (\nabla\sigma)^2] \tag{1}$$

$$- \frac{a^4 m^2}{2}\sigma^2 + a^3 \rho q'\sigma \Big\}. \tag{2}$$

We use $q(\boldsymbol{x}) = \int \mathrm{d}^3k\, q_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{x}}/(2\pi)^3$ and define P_q and B_q after removing the momentum-conserving $(2\pi)^3\delta^3$ factors. The dimension-one field $q = C\zeta$ has an unspecified conversion C to curvature; we keep this normalisation explicit. The single-clock curvature reference follows from the dilation response $-(3 + k\partial_k)P_\zeta$, with $P_\zeta \propto k^{n_s-4}$:

$$\lim_{k_L/k_S\to 0} \frac{B_\zeta}{P_\zeta(k_L)P_\zeta(k_S)} = 1 - n_s = 0.035 \quad (n_s = 0.965). \tag{3}$$

This specified tilted reference is distinct from the exact de Sitter spectator calculation. It is not inserted into the computed spectator bispectrum.

Step contributions: Priyansh Rawat: Physical setting, quadratic action and shared conventions; review of numerical checks and interpretation; **Samarth Bhatnagar**: Fourier conventions, dimensional analysis and the q–ζ conversion; review of the squeezed-limit derivation; **Mayank Sinha**: Numerical implementation, validation, data and figures; reproduction instructions and report consistency.

Step 2: Two-oscillator Gaussian benchmark

The two-oscillator ground state with $K = \left(\begin{smallmatrix} 1 & 0.3 \\ 0.3 & 2 \end{smallmatrix}\right)$ has $X = K^{-1/2}/2$ and $P = K^{1/2}/2$. The reduced eigenvalue $\nu = \sqrt{X_{11}P_{11}}$ gives

$$s(\nu) = (\nu + \tfrac{1}{2})\ln(\nu + \tfrac{1}{2}) - (\nu - \tfrac{1}{2})\ln(\nu - \tfrac{1}{2}). \tag{4}$$

The benchmark is $\nu = 0.5028165419863$, $s = 0.0193599304236$ nats. Zero coupling gives zero entropy; increasing the coupling to 0.6 gives $s = 0.06727111385$. Local canonical rescaling leaves the determinant unchanged to 5.6×10^{-17}. Direct Gaussian-kernel purity, $\mathrm{Tr}\rho_1^2 = 1/(2\nu)$, checks the covariance calculation independently. For general coupling, $d = \sqrt{2 - c^2}$ gives $\nu(c) = \sqrt{(2+d)(1+d)/[4d(3+2d)]}$ and the exact curve $S(c) = s(\nu(c))$. The Step 2 plot overlays this curve with the Gaussian-kernel entropy from eigenvalues $(1-\xi)\xi^n$, $\xi = (1-\mathcal{P})/(1+\mathcal{P})$. The two agree within 10^{-12} nats across 701 couplings $0 \le c \le 1.4$.

Step contributions: Priyansh Rawat: Normal-mode derivation and benchmark review; **Samarth Bhatnagar**: Reduced-state covariance and rescaling checks; **Mayank Sinha**: Numerical diagonalisation, purity cross-check and figure.

Step 3: Canonical dynamics and independent equations

Derivative mixing changes the momentum: $\pi_q = a^2 q' + a^3 \rho\sigma$. The Legendre transform is

$$H_k = \frac{(\pi_q - a^3\rho\sigma)^2 + \pi_\sigma^2}{2a^2} + \frac{a^2k^2}{2}(q^2 + \sigma^2) + \frac{a^4m^2}{2}\sigma^2. \tag{5}$$

The ρ^2 term is essential. Two complex mode columns are evolved using canonical equations and, independently, velocity equations in $N = -\ln x$:

$$\ddot{Q} + 3\dot{Q} + x^2 Q = -g(\dot{\Sigma} + 3\Sigma), \tag{6}$$

$$\ddot{\Sigma} + 3\dot{\Sigma} + (x^2 + M^2)\Sigma = g\dot{Q}, \tag{7}$$

$$\Pi_q = (\dot{Q} + g\Sigma)/x^3, \quad \Pi_\sigma = \dot{\Sigma}/x^3. \tag{8}$$

Identical finite-time initial data give a maximum scaled column difference of 2.010×10^{-7} at the initial comparison tolerance. The symbolic Legendre-transform identity is exact. Production tests then use tighter tolerances and controlled initial states. For $V = \mathrm{Re}(\mathcal{M}\mathcal{M}^\dagger)$, the full state has symplectic eigenvalues $1/2$, while the light subblock can have $\nu_q > 1/2$. Entropies are per real sector; the identical cosine and sine sectors of a nonzero $\pm\boldsymbol{k}$ pair contribute twice this entropy.

Step contributions: Priyansh Rawat: Legendre transform and Hamiltonian derivation; **Samarth Bhatnagar**: Dimensionless and velocity equations; convention checks; **Mayank Sinha**: Independent numerical formulations and agreement tests.

Step 4: Free modes and the ultraviolet state

The uncoupled light mode is $Q = (1 - ix)e^{ix}/\sqrt{2}$. The heavy mode is

$$\Sigma = \frac{\sqrt{\pi}}{2} e^{i\pi(\nu/2+1/4)} x^{3/2} H_\nu^{(1)}(x), \qquad \nu = \sqrt{9/4 - M^2}. \tag{9}$$

Complex-order Hankel evaluation and its derivative reproduce both fields with scaled errors below 6×10^{-10} in the free test. The light power approaches $1/2$.

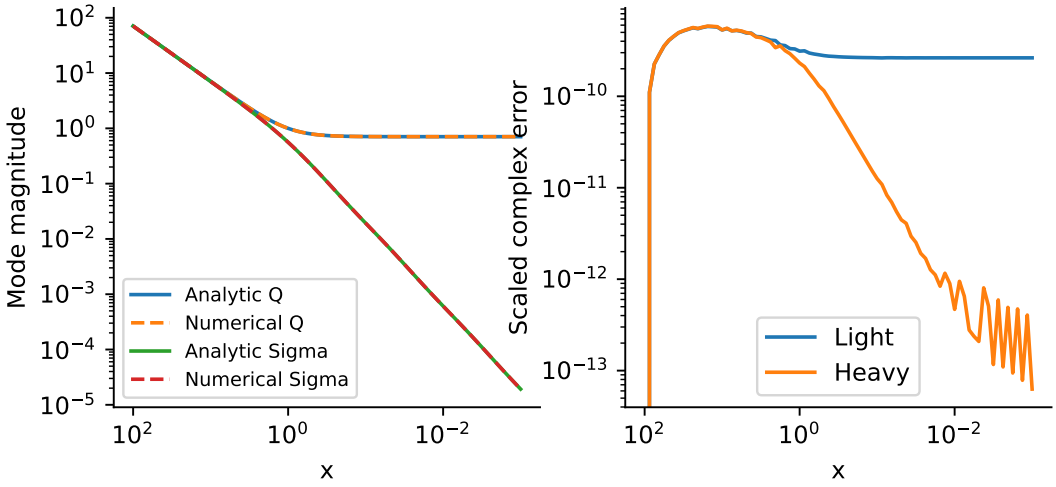


Figure 1: Free analytic and numerical modes agree in magnitude and complex phase.

A product vacuum imposed at finite X gives visible initial-state contamination in the much smaller bispectrum. We instead construct a positive-frequency ultraviolet state through $Y' = R(x)Y$, $Y = (Q, \Sigma)^T$, with $R = iI + \sum R_n/x^n$. Its Riccati equation follows directly from the coupled field equations. With $J_{12} = 1$, $J_{21} = -1$ and $D_{12} = -3g$, $D_{22} = -M^2$,

$$R_1 = I + gJ/2, \quad 2iR_n = [(n+1)I + gJ]R_{n-1} - \sum_{j=1}^{n-1} R_j R_{n-j} + \delta_{n2} D. \tag{10}$$

At X, choose the position covariance $FF^T = X^2(\mathrm{Im}\, R)^{-1}/2$ and velocity columns $-XRF$. This imposes canonical initial commutators. Comparing orders $4/6$ and $X = 100, 200, 400, 800$ verifies the ultraviolet limit. No correction is imposed on the evolved covariance.

Step contributions: Priyansh Rawat: Bunch–Davies modes and initial-state conventions; **Samarth Bhatnagar**: Canonical commutator and starting-time checks; **Mayank Sinha**: Free-mode implementation, adiabatic tests and figures.

Step 5: Gaussian entropy and power

The benchmark uses sixth-order initial data, $X = 400$, relative/absolute tolerances $2.5 \times 10^{-14}/10^{-17}$ and the local shear $\widetilde{\Pi}_q = \Pi_q + Q/x$. Extended-precision covariance products reduce late-time cancellation. Samples require commutator residual $< 10^{-6}$ and full-state eigenvalue error $< 10^{-5}$.

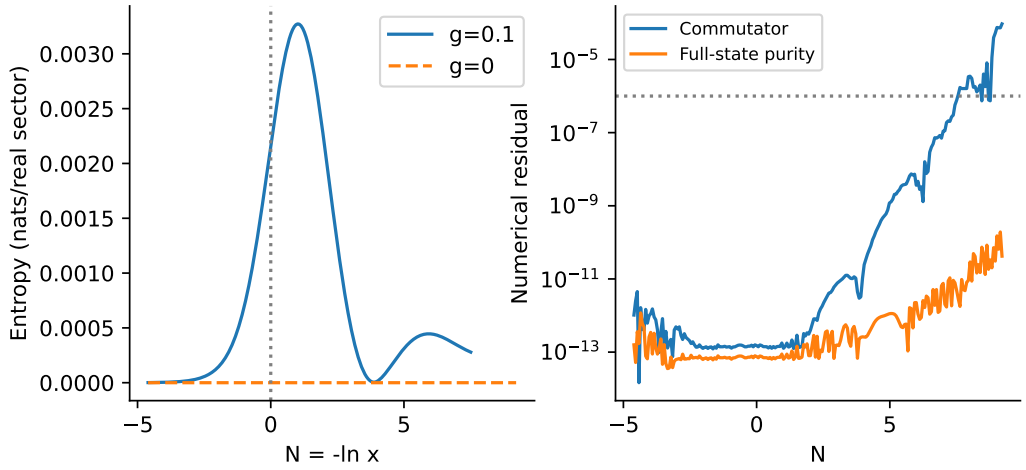


Figure 2: Entropy history and purity/commutator diagnostics. The curve stops at the first failure of the numerical criteria.

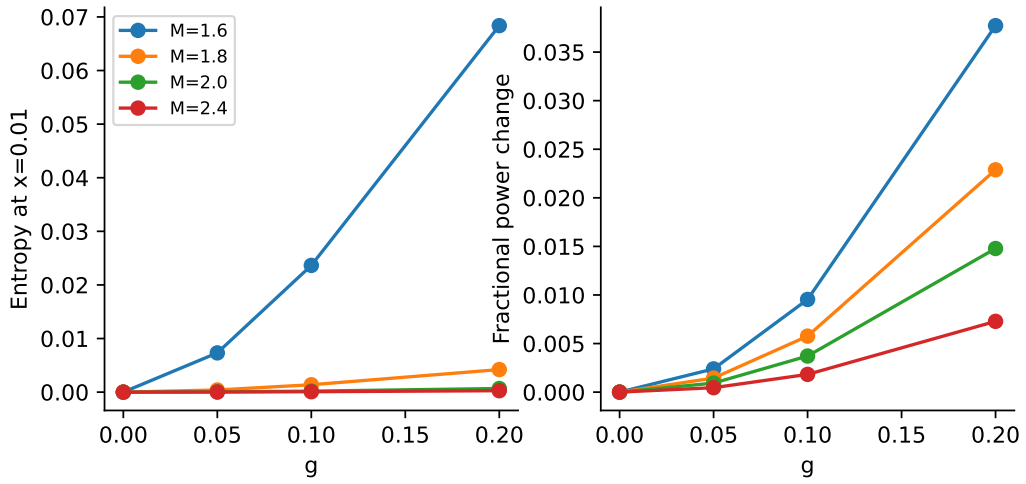


Figure 3: Mass and mixing comparison at $x = 0.01$; entropy and power change are separate diagnostics.

All 64 requested mass/mixing/time comparison rows pass. At $M = 2$, $g = 0.1$, $x = 0.01$, $S_q = 1.845987 \times 10^{-4}$ nats and $\Delta P_q/P_{q,0} = 0.0037140$. The zero-mixing history has numerical entropy below 2.3×10^{-9}; reversing g changes the benchmark entropy by 6.1×10^{-12}. Non-monotone entropy evolution is compatible with unitary evolution of the joint pure state. The last reliable contiguous history sample is $x \simeq 5.49 \times 10^{-4}$.

Step contributions: Priyansh Rawat: Covariance and entropy derivation; interpretation; **Samarth Bhatnagar**: Purity, zero-mixing and sign-symmetry validation; **Mayank Sinha**: Mass/mixing scan, reliability mask and figures.

Step 6: Heavy-field clock and independent in-in benchmark

Outside the horizon the heavy branch behaves as $x^{3/2\pm i\mu}$ with $\mu = \sqrt{M^2 + g^2 - 9/4}$. At $M = 2$, $g = 0.1$, this gives $\mu = 1.326649916$. This exponent sets the logarithmic clock frequency tested in Step 9.

At first order in the interaction Hamiltonian,

$$\Delta\langle O\rangle = -i \int \mathrm{d}\eta\, \langle [O, H_I]\rangle_0. \tag{11}$$

An independent sine quadrature checks the massless test Hamiltonian $H_I = \gamma \int a(q')^3/3!$. Its analytic result is

$$B_{\text{test}} = -\frac{\gamma H^5}{2k_1k_2k_3(k_1+k_2+k_3)^3}. \tag{12}$$

The regulated integral is $2/[(\epsilon+i)K]^3$. Equilateral and squeezed tests at $\epsilon = 10^{-3}$ agree with the zero-regulator answer to 6.0×10^{-6} relative, meeting the 0.1% requirement.

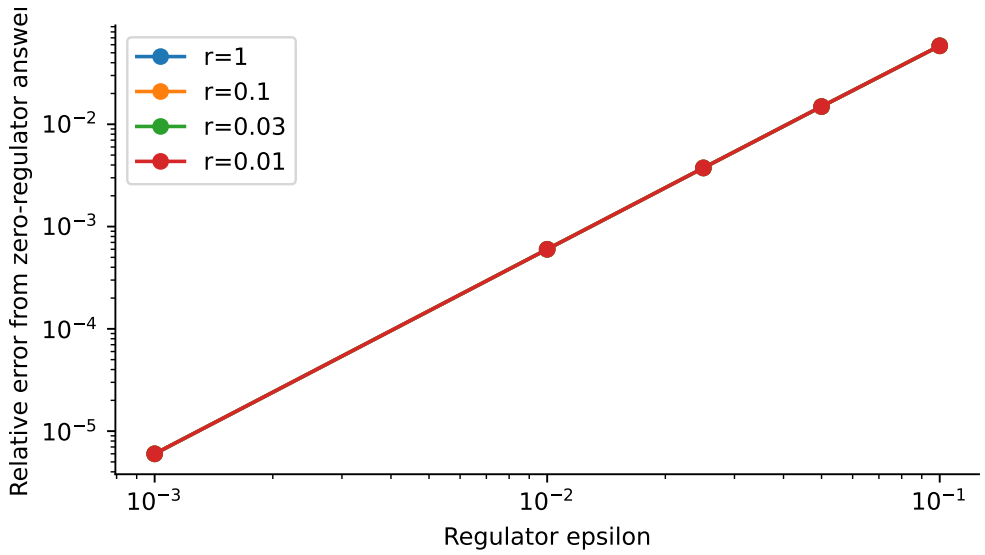


Figure 4: Independent oscillatory quadrature reproduces the analytic test and its regulator dependence.

Step contributions: Priyansh Rawat: Heavy-clock exponent and in-in derivation; **Samarth Bhatnagar**: Sign and contraction-factor checks; benchmark review; **Mayank Sinha**: Independent oscillatory quadrature and regulator tests.

Step 7: Mixed contractions and the spectator bispectrum

For $H_I = \lambda_3 \int a^4\sigma^3/3!$, define $K_k = \sum_a q_k^a(\eta_f)\sigma_k^{a*}(\eta)$. The six contractions cancel 3! and the commutator gives

$$B_q = 2\lambda_3 \int \mathrm{d}\eta\, a^4\, \mathrm{Im}(K_1K_2K_3), \quad F(r) = \frac{HB_q(r,1,1)}{P_q(r)P_q(1)}. \tag{13}$$

At a common physical final time the long leg ends at rz_f, not z_f. We keep the complex contractions, mode columns and cumulative integrals. The $g = 0$ and $\lambda_3 = 0$ controls vanish exactly; doubling λ_3 doubles F, and weak mixing gives the expected g^3 scaling.

Step contributions: Priyansh Rawat: Mixed-contraction and bispectrum derivation; **Samarth Bhatnagar**: Common-final-time and zero/cubic controls; **Mayank Sinha**: Complex-mode integration, saved phases and figures.

Step 8: Convergence of the baseline physical signal

At $M = 2$, $g = 0.1$, $\lambda_3/H = 0.01$, production integration uses $X = 1600$ and sixth-order ultraviolet data. We test regulator/cutoff pairs $(0.1, 80), (0.05, 160), (0.025, 320)$ at doubled cutoffs, add $\epsilon = 0.0125$, and extrapolate the last three regulators with linear and quadratic models. Early intervals resolve oscillations; late integration uses $\ln z$. Grid refinement and independent adaptive quadrature agree at the $r = 0.03$ benchmark to 1.7×10^{-22} in regulated F. Initial X, ODE precision, quadrature, regulator, early tail and final time are varied separately. The sum of their changes is a conservative numerical budget, not a statistical error bar.

r	F extrapolated	Numerical budget
0.2	1.812×10^{-8}	1.059×10^{-10}
0.1	1.585×10^{-9}	2.857×10^{-11}
0.03	-3.986×10^{-10}	1.104×10^{-12}
0.01	6.258×10^{-11}	2.940×10^{-13}
0.003	2.482×10^{-11}	4.818×10^{-14}
0.001	-2.966×10^{-13}	1.414×10^{-14}

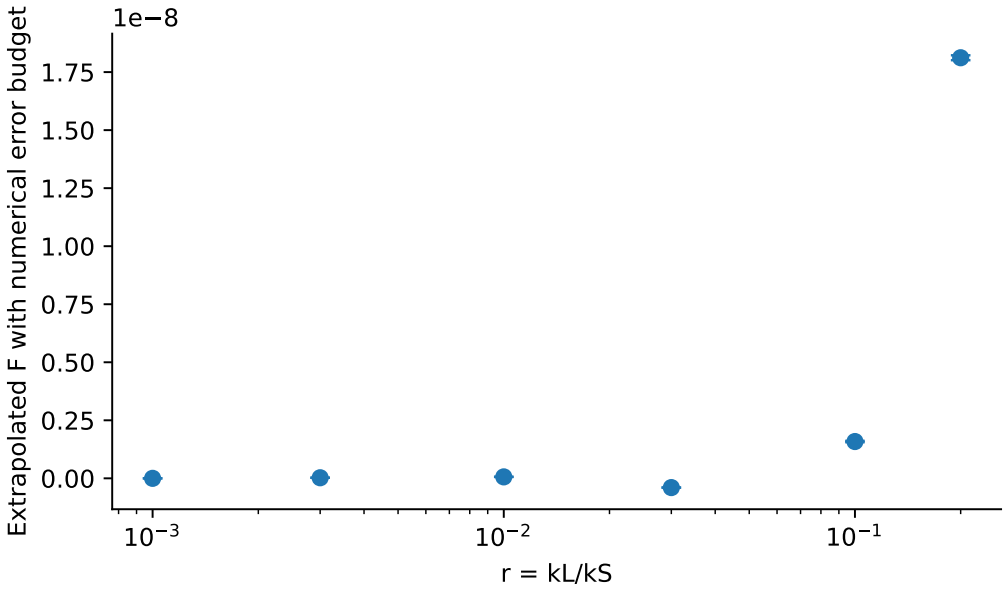


Figure 5: Physical pilot values and the tested numerical budget. Near a clock zero, relative error in the total value can be large while envelope precision remains useful.

The largest pilot error normalised to a clock-envelope scale is 0.00585. In the 50-point shape scan the largest numerical error is less than 0.009 of the fitted oscillatory envelope. This supports studying the resolved shape while still requiring the fit-window checks below.

Step contributions: Priyansh Rawat: Ultraviolet initial-state derivation and interpretation; **Samarth Bhatnagar**: Regulator, cutoff and final-time error budget; **Mayank Sinha**: Quadrature/precision comparisons and convergence data.

Step 9: Mass recovery and physical fit stability

9.1 Mock recovery and no-clock controls

Using the heavy-field frequency derived in Step 6, we profile frequency with

$$F = b_0 + b_2 r^2 + r^{3/2}[A\cos(\mu \ln r) + B\sin(\mu \ln r)], \tag{14}$$

solving linearly for nuisance coefficients at each frequency. The scan covers $0.1 \leq \mu \leq 4$ in steps 0.01, refining every candidate minimum. The mocks use $M = 2$, $g = 0.1$, $\mu = 1.326649916$, $A = 0.1$, $B = 0.05$, $b_0 = 0$, $b_2 = 0.2$, 50 logarithmic momenta and known independent noise. There are 100 trials at each of 1% and 5% oscillatory-RMS noise; the seed is 20261029.

Noise	Frequency RMSE	95% coverage	Held-out null false alarm
1%	0.04687	93%	2%
5%	0.205	94%	4%

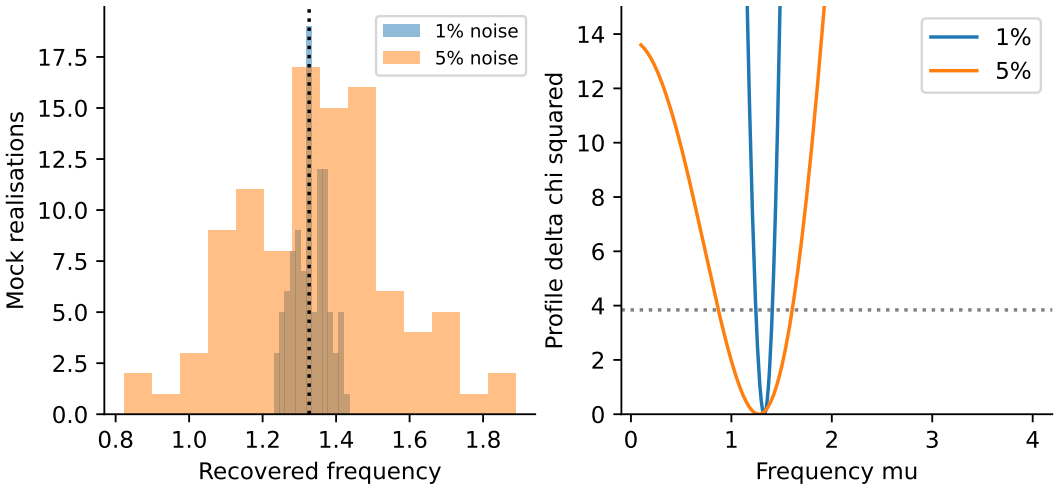


Figure 6: Mock recovered frequencies and profile curves. The likelihood intervals apply to these known Gaussian-noise experiments.

The no-clock thresholds are calibrated on separate null samples and tested on 100 independent null realisations per noise level. These rates have sampling uncertainty of a few percentage points. They explicitly include the search over frequency, which is not identifiable under the no-clock hypothesis.

9.2 Physical mass assessment

The converged physical scan uses $10^{-3} \leq r \leq 0.2$. Its numerical errors are used as fit weights, with background and endpoint variations assessed separately. The fitted full-range coefficient amplitude is approximately 1.389×10^{-7}, with only 1.11 visible cycles. We convert frequency with

$$m/H = \sqrt{\mu^2 + 9/4 - g^2}. \tag{15}$$

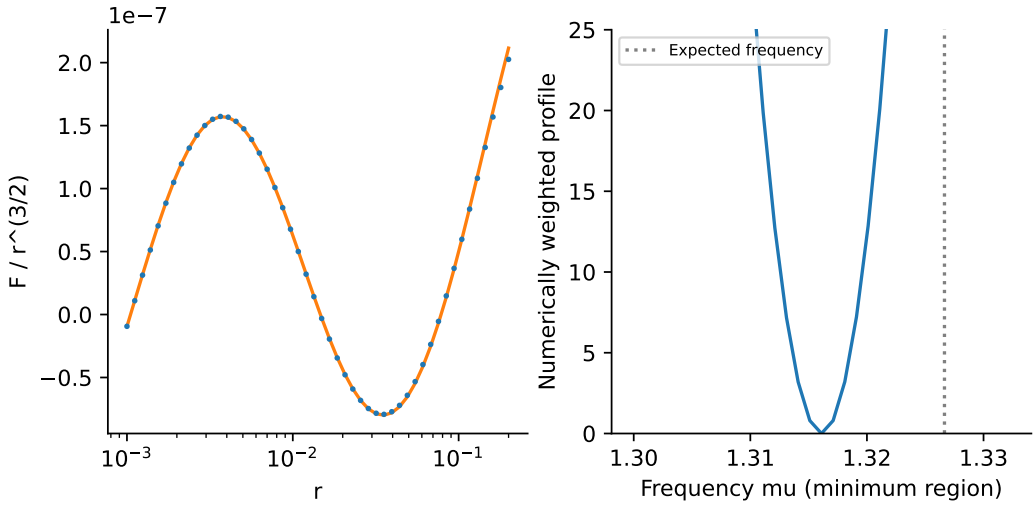


Figure 7: Physical squeezed shape and its numerical frequency profile. The narrow profile under one model excludes neither background nor endpoint sensitivity.

$r_{\max}$	Extra r^4	μ	m/H
0.05	no	1.32317	1.99769
0.05	yes	1.32544	1.9992
0.1	no	1.32041	1.99587
0.1	yes	1.32445	1.99854
0.2	no	1.3161	1.99302
0.2	yes	1.32196	1.99689

Including both lower endpoints, $r_{\min} = 0.001, 0.003$, gives the best-fit frequency envelope 1.3161–1.3564 and mass envelope 1.9930–2.0198. This is model/window sensitivity, not a confidence interval with demonstrated coverage. It contains the input mass $M = 2$, but does not justify reporting a highly precise mass from a single narrow profile. If mixing is unknown, the clock identifies the mass–mixing combination rather than m alone.

Step contributions: Priyansh Rawat: Frequency model and mass-conversion interpretation; **Samarth Bhatnagar**: Mock/no-clock calibration and fit-window review; **Mayank Sinha**: Profile fitting, physical shape scan and figures.

Step 10: Entropy, cubic coupling and conclusions

Changing λ_3 at fixed M, g leaves the Gaussian entropy unchanged and scales the leading bispectrum. The same quadratic entropy therefore accompanies zero, half-strength and full-strength three-point signals.

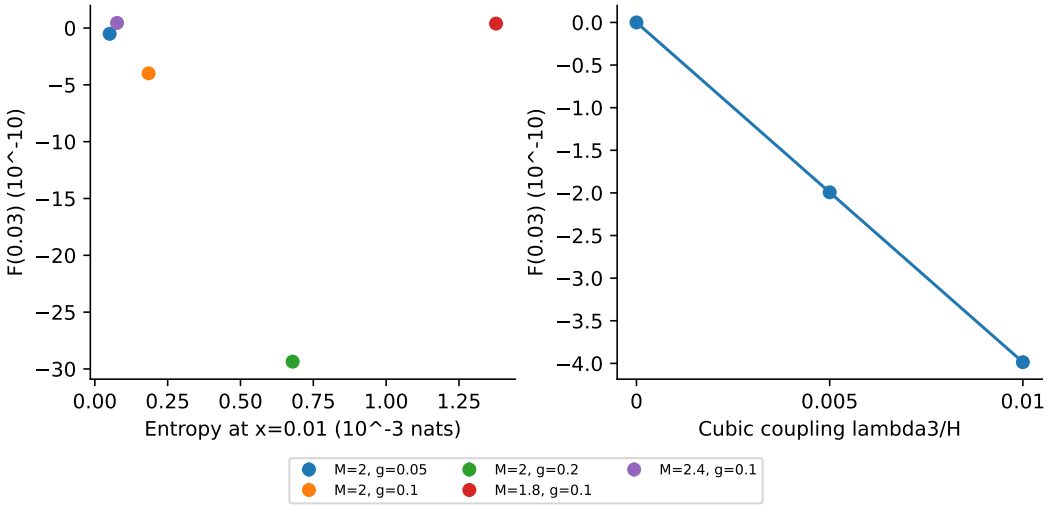


Figure 8: Selected entropy–signal points and the independent cubic-coupling control. A single triangle also contains phase information; selected-point bars show regulator variation.

λ_3/H	S_q	$F(0.03)$
0	1.846×10^{-4}	0
0.005	1.846×10^{-4}	-1.993×10^{-10}
0.01	1.846×10^{-4}	-3.986×10^{-10}

This establishes that Gaussian entropy alone cannot determine the spectator three-point amplitude. A fixed-coupling correlation study can still be meaningful, but must hold time, numerical accuracy and measurement covariance fixed. An illustrative fixed covariance $\sigma_F^2 I$, $\sigma_F = 10^{-10}$, makes known-template signal/noise proportional to the cubic coupling; equal fractional noise would conceal that dependence. The core calculation has validated entropy, an independent in-in benchmark, a converged baseline bispectrum and an explicit mass-fit stability assessment. Late-time entropy is limited by covariance conditioning; physical mass precision is limited by the available oscillation range and background model. Broader physical amplitude scans are optional extensions. No survey detectability or Bell-test claim follows from these model calculations. All scripts, inputs, essential tables, complex-mode archives, LaTeX sources and PDF reports are organised by step in the linked repository. Run `python3 run_steps.py` after installing the pinned requirements; compile each report from its `latex/` directory.

Step contributions: **Priyansh Rawat**: Entropy–signal comparison and conclusions; **Samarth Bhatnagar**: Cubic-coupling and common-covariance controls; **Mayank Sinha**: Selected parameter comparisons, numerical tables and figure.

References

[1] C. Weedbrook et al., *Gaussian Quantum Information*, Rev. Mod. Phys. 84, 621 (2012), arXiv:1110.3234.

[2] X. Chen and Y. Wang, *Quasi-Single Field Inflation and Non-Gaussianities*, JCAP 04 (2010) 027, arXiv:0911.3380.

[3] N. Arkani-Hamed and J. Maldacena, *Cosmological Collider Physics*, arXiv:1503.08043.

Code and supporting data: final_report on GitHub.